KAUST Mathematics Competition (KMC): Regional Qualifier

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KAUST Mathematics Competition (KMC): Regional Qualifier Practice Exercises

1 [KAUST KMC — Regional Qualifier] By grouping consecutive perfect-square intervals along $y = \sqrt{x}$ shown in the plot, compute the exact value of $\sum_{k=1}^{15} \lfloor \sqrt{k} \rfloor$.

a. 38

b. 14

c. 34

d. 31

2 [KAUST KMC — Regional Qualifier] By grouping consecutive perfect-square intervals along $y = \sqrt{x}$ shown in the plot, compute the exact value of $\sum_{k=1}^{24} \lfloor \sqrt{k} \rfloor$.

a. 75

b. 70

c. 30

d. 66

3 [KAUST KMC — Regional Qualifier] By grouping consecutive perfect-square intervals along $y = \sqrt{x}$ shown in the plot, compute the exact value of $\sum_{k=1}^{35} \lfloor \sqrt{k} \rfloor$.

a. 125

b. 131

c. 55

d. 120

4 [KAUST KMC — Regional Qualifier] By grouping consecutive perfect-square intervals along $y = \sqrt{x}$ shown in the plot, compute the exact value of $\sum_{k=1}^{48} \lfloor \sqrt{k} \rfloor$.

a. 210

b. 91

c. 197

d. 203

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5 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a **monic** polynomial of degree 3 such that $P(k) = 2k$ for every integer $k \in \{1, 2, \ldots, 3\}$. Find the exact value of $P(4)$ (note the classic linear-extrapolation trap $8!$).

a. 8

b. 6

c. 14

d. 12

6 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a **monic** polynomial of degree 4 such that $P(k) = 2k$ for every integer $k \in \{1, 2, \ldots, 4\}$. Find the exact value of $P(6)$ (note the classic linear-extrapolation trap $12!$).

a. 12

b. 132

c. 120

d. 32

7 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a **monic** polynomial of degree 5 such that $P(k) = 2k$ for every integer $k \in \{1, 2, \ldots, 5\}$. Find the exact value of $P(6)$ (note the classic linear-extrapolation trap $12!$).

a. 132

b. 12

c. 120

d. 130

8 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a **monic** polynomial of degree 3 such that $P(k) = 3k$ for every integer $k \in \{1, 2, \ldots, 3\}$. Find the exact value of $P(5)$ (note the classic linear-extrapolation trap $15!$).

a. 15

b. 24

c. 40

d. 39

9 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a polynomial of degree **at most** 2 satisfying $P(k) = \frac{24}{k}$ for $k = 1, 2, \ldots, 3$. Compute the exact value of $P(4)$.

a. 6

b. 16

c. 12

d. 13

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10 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a polynomial of degree **at most** 3 satisfying $P(k) = \frac{120}{k}$ for $k = 1, 2, \ldots, 4$. Compute the exact value of $P(5)$.

a. 24

b. 0

c. 5

d. 48

11 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a polynomial of degree **at most** 4 satisfying $P(k) = \frac{720}{k}$ for $k = 1, 2, \ldots, 5$. Compute the exact value of $P(6)$.

a. 240

b. 120

c. 246

d. 241

12 [KAUST KMC — Regional Qualifier] Let $P(x)$ be a polynomial of degree **at most** 5 satisfying $P(k) = \frac{5040}{k}$ for $k = 1, 2, \ldots, 6$. Compute the exact value of $P(7)$.

a. 720

b. 7

c. 1440

d. 0

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